

Free Energy

An explanation of Karl Friston's Free Energy Principle (FEP) — already name-checked in the *Cognitive Systems Engineering* book sitting in this repo (Chapter 30, tied to Piaget's assimilation/accommodation and Boyd's OODA loop).

The Problem It's Trying to Solve

Any self-organizing system that persists over time — a cell, an organism, a brain — has to occupy a limited, low-entropy set of states (the right temperature, hydration, blood glucose, etc.) against the universe's general pull toward disorder. Persisting means resisting dissipation. FEP asks: what must such a system be doing, computationally, to pull this off?

The Core Move

The most direct answer would be "minimize surprise" — avoid sensory states that are improbable given your model of the world (formally, surprisal = $-\log P(\text{sensory input} \mid \text{model})$). But computing true surprise exactly is intractable — it would require marginalizing over every possible hidden cause of your sensory input. So instead, Friston proposes the system minimizes a computable upper bound on surprise called **variational free energy**. This isn't the same "free energy" from thermodynamics (Gibbs/Helmholtz) — it's a borrowed term from variational Bayesian statistics (the same math underlies things like variational autoencoders in ML).

Free energy can be decomposed two useful ways:

- **Complexity – Accuracy**: a Bayesian-Occam's-razor tradeoff between how far your beliefs deviate from a simple prior and how well those beliefs predict the data.
- **Surprise + KL divergence** (between your approximate internal model and the true posterior): since KL divergence is always ≥ 0 , free energy is always $\geq$ surprise. Minimize the bound, and you drag down the actual surprise too — and as your internal model gets more accurate, the bound tightens toward the true value.

Two Ways to Minimize It — the Key Unifying Move

1. **Perceptual inference**: update your internal beliefs to better explain the sensory data you're getting (classic Bayesian updating / predictive coding — perception as hypothesis-revision).
2. **Active inference**: instead of updating your beliefs, act on the world to change the sensory data so it matches your existing predictions.

That's the theory's most distinctive claim: perception and action aren't two separate systems bolted together — they're the *same* optimization (minimizing free energy) run in two directions. You can resolve a mismatch

between model and world either by changing the model or by changing the world.

Hierarchical Structure

The brain is modeled as a multi-level generative model constantly issuing top-down predictions; prediction errors flow upward and get absorbed either by updating beliefs or by triggering action, at every level simultaneously, each level operating over a different timescale.

The Connection Back to This Repo

Chapter 30 of the CSE book explicitly lines up FEP with Piaget's assimilation (acting on the world to make it fit your schema — active inference) and accommodation (updating the schema to fit surprising feedback — perceptual inference), and with Boyd's creation/destruction dynamic in the OODA loop. It's the same move as Rasmussen's Abstraction-Decomposition hierarchy, too — higher levels provide context that filters what's salient at lower levels, exactly FEP's hierarchical top-down/bottom-up structure. And it's a formal, mechanistic answer to the question the "Three classes of systems" note raises about self-organizing vs. self-designing systems: FEP is Friston's attempt to give a rigorous account of *how* a self-designing system (one with "internal agency") actually manages the trick of persisting against entropy by acting on top of just dissipating energy efficiently.

A Caveat Worth Knowing

FEP is contested. Critics argue it's so general it risks being unfalsifiable — that it can be retrofitted to explain almost any behavior after the fact — and that it's substantially a reformulation of older ideas (Helmholtz's "unconscious inference," the Bayesian brain hypothesis, homeostasis/allostasis, classical cybernetics) in new mathematical clothing rather than a novel empirical discovery. That doesn't make it wrong, but it's not settled, universally-accepted science the way the label "principle" might suggest.